Customer Churn Prediction Analysis Business Analyst Internship Project

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INTRODUCTION

- Customer churn refers to customers discontinuing a service.
- Churn affects company revenue and growth.
- Predicting churn helps retain valuable customers.
- Data analytics enables proactive decision-making.
- Machine learning can identify potential churners.

PROBLEM STATEMENT

- High churn rates reduce business profitability.
- Customer retention is more cost-effective than acquisition.
- Businesses need early identification of at-risk customers.
- Manual prediction methods are inefficient.
- A data-driven solution is required for churn management.



OBJECTIVES

- Analyze customer behavior and patterns.
- Identify factors influencing customer churn.
- Build a predictive machine learning model.
- Evaluate model performance using key metrics
- Provide actionable business recommendations.



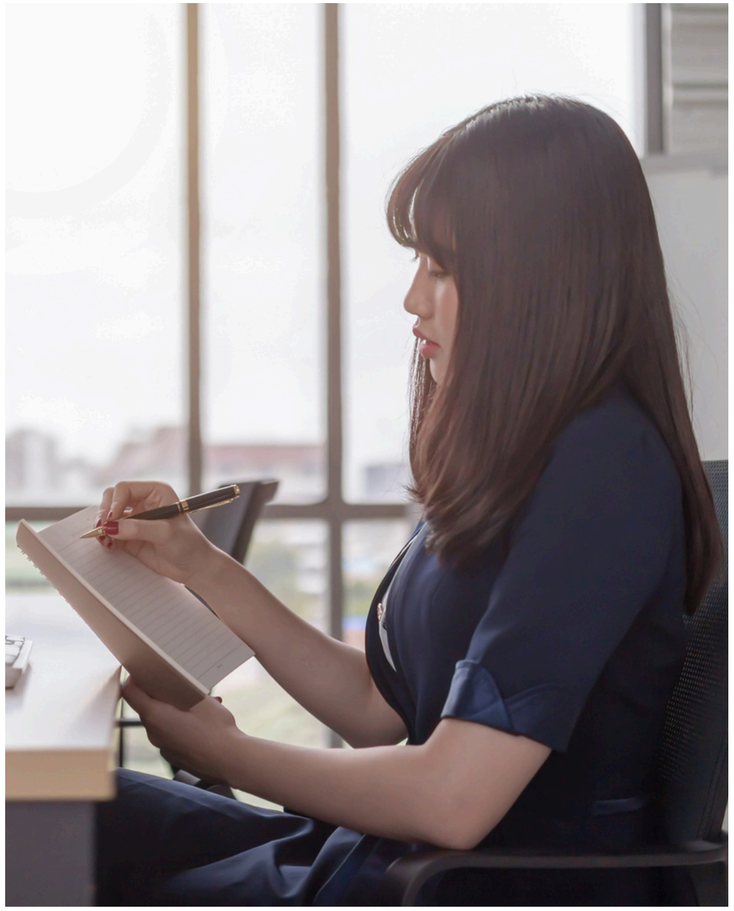
DATASET OVERVIEW

- Dataset contains customer demographic information.
- Includes service usage and billing details.
- Features such as tenure and monthly charges are analyzed.
- Target variable indicates churn status (Yes/No).
- Data is used for predictive modeling.



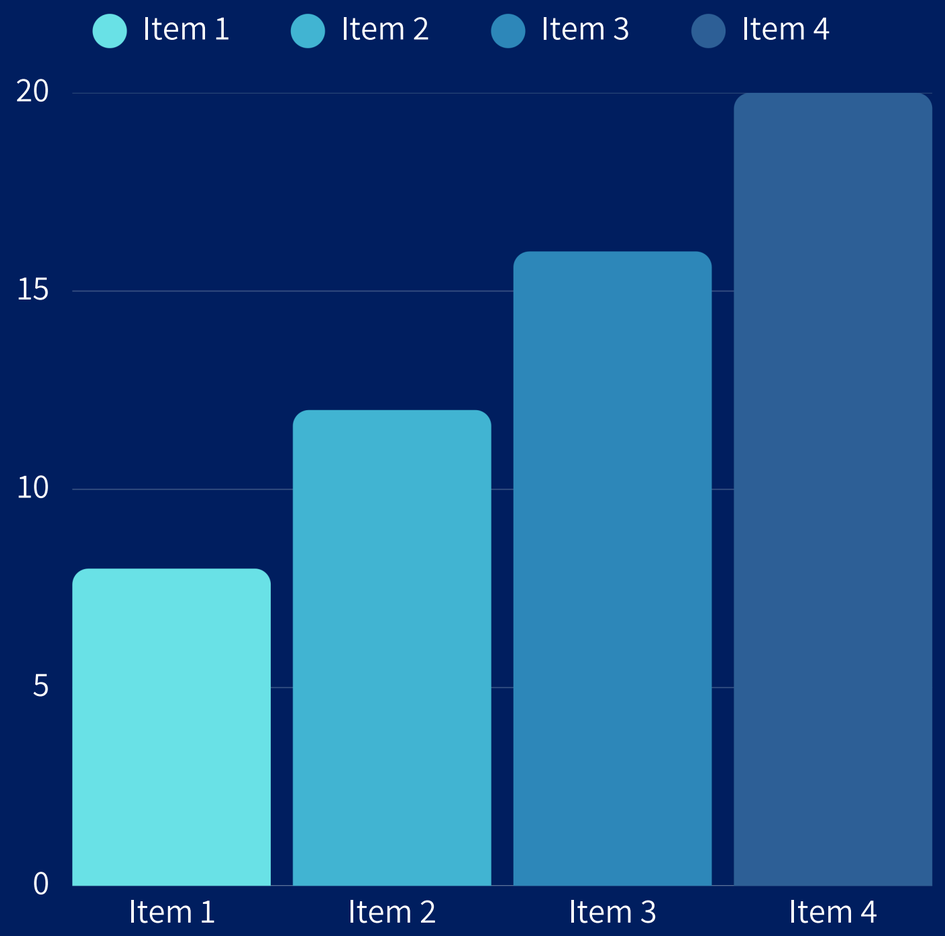
DATA PREPROCESSING

- Checked and handled missing values.
- Removed duplicate records if present.
- Encoded categorical variables into numerical format.
- Standardized and cleaned the dataset.
- Prepared data for machine learning algorithms.



EXPLORATORY DATA ANALYSIS

- Analyzed churn distribution among customers.
- Examined relationship between tenure and churn.
- Compared monthly charges across customer groups.
- Studied impact of contract types on churn.
- Identified patterns and trends in customer behavior.



MODEL BUILDING

- Split data into training and testing sets.
- Applied machine learning algorithms.
- Trained models on historical customer data.
- Evaluated performance using accuracy and recall.
- Selected the best-performing model.



FEATURE IMPORTANCE ANALYSIS

- Identified the most influential churn factors.
- Contract type significantly impacts retention.
- Customer tenure affects churn probability.
- Monthly charges influence customer decisions.
- Feature importance helps target retention efforts.



CONCLUSION & RECOMMENDATIONS

- Churn prediction model successfully identified risk factors.
- Customer behavior patterns were effectively analyzed.
- Long-term contracts can reduce churn rates.
- Personalized offers may improve retention.
- Data-driven strategies support business growth.



🙏THANK YOU

Questions & Discussion

📊 Customer Churn Prediction Analysis

“Turning customer data into actionable business insights.”

